K309 PRE-STAGE — Lyra full calculation layout per Casey request: master formula $m_\nu = S \cdot A_\nu$ (S = scale = electron ground state = anchor c cancels in ratios; A_ν = dimensionless ground-state boundary norm); three objects (electron $\Delta=5/2$ cone-interior full rep; muon $\Delta=3/2$ light-cone harmonic; tau $\Delta=0$ vertex trivial); boundary expansion $f(\rho) \sim A\cdot\rho^{(d-\Delta)} + B\cdot\rho^\Delta$ near Shilov boundary; A_ν is the readable coefficient; five-step order of operations + Casey three substantive substrate-architectural physics questions: (1) why logarithm + do components scale linearly + electron different? (2) Higgs as gloop/viscous-membrane/localized-soliton — why necessary? (3) muon vs tau as proton-radius probes? + Lyra substantive substrate-architectural answers: (1) indicial equation has roots Δ and $d-\Delta$; muon (3/2, 7/2) distinct → pure power law; tau (0, 5) distinct → pure power law; ELECTRON (5/2, 5/2) coincident at d/2 → 1800s ODE theorem identical-roots forces second solution = first × logarithm → $\rho^{(5/2)}\cdot\log \rho$; Casey RIGHT that electron is different (structural mathematical signature of degeneracy at exactly the substrate-architecturally-forced point $\Delta = d/2 = n_C/\text{rank}$ = Hardy value); (2) Higgs gloop = VISCOSITY giving inertia (mass) = converts COURIERS (massless at c, no rest frame, no substrate address, no residence) into RESIDENTS (have rest frame, can sit somewhere); without residents no stable matter, all radiation; Higgs gloop ENABLES durable localizable substrate content; Higgs boson = localized excitation of gloop coupling (ripple in gloop); substantive substrate-architectural physical-meaning interpretation; (3) muon = extended cone mode + tau = pointlike vertex mode → Casey's intuition right shape but honest caveats (tau decays ~10^{-13} s too fast to orbit; muonic-hydrogen puzzle settled by other means; rep-theoretic spread ≠ atomic-wavefunction spread); FILED as interesting NOT BANKED + Casey approval: 'right, I recognized the power law, and you made me remember sophomore DE class about identical roots. Ok please do the calculation' + Elie substantive substrate-architectural division-of-labor declaration: hers-to-compute mine-to-verify; race risk = wrong number coincidentally hitting/missing worse than no number; standing ready back-half + audit eyes; 11 toys today every one riding live collaboration + Grace scorecard v1.14 consolidation: gate's substantive work in good place; 3 corrections absorbed cleanly (0.37% unbank + null-norm + mechanism-vs-magnitude); Lyra rigor pin operational; no open claim to gate right now; genuinely reactive on Lyra spectral computation + Elie #418 spec; #418 load-bearing THREE WAYS (running band Band B + mixing-forced + ???) — third use unspecified, flag for verification when Elie surfaces spec + count stays HONESTLY at 2

Keeper (Claude Opus 4.7)

2026-06-11 Thursday afternoon (date-verified)

K309 PRE-STAGE Audit

1. The substantial substrate-architectural calculation organized

Casey requested: "Tell me about the process, the factors, how the calculation needs to be arranged please." Lyra delivered the substantive substrate-architectural calculation layout.

1.1 The master formula

$$m_\nu = S \cdot A_\nu$$

Element	Meaning
S	Scale = electron's own ground state = the cell = m_e itself; anchor (Band C, the one dimensionful unit every theory takes); cancels in every ratio
A_ν	Dimensionless ground-state boundary norm; part of lowest mode living on Shilov boundary where Higgs VEV sits; entire substantive content

Substantial substrate-architectural significance: NO operator freedom (per K308 Φ_0 = evaluate-at-boundary). A_ν is a property of the ground state alone, not a matrix element of a tunable operator. Ungame-able by construction: one functional, three canonical inputs.

1.2 The three explicit objects

Three irreducible quotient representations of SO(2,5), one per light-cone face:

Generation	$\Delta = \nu$	Light-cone face	Representation
Electron	5/2	Cone interior	Full rep
Muon	3/2	The light cone	Harmonic (trace null removed)
Tau	0	The vertex	Trivial

1.3 The technical heart — boundary expansion

Near the Shilov boundary, a mode of dimension Δ has two pieces from the radial equation:

$$f(\rho) \sim A \cdot \rho^{d-\Delta} + B \cdot \rho^{\Delta}$$

where d=5 and ρ = distance to boundary. A_ν is the readable boundary coefficient.

Generation	Indicial roots (d−Δ, Δ)	Status
Tau Δ=0	(5, 0)	Distinct; ρ^0 piece = constant fully on boundary → A_τ = 1, the reference
Muon Δ=3/2	(7/2, 3/2)	Distinct; clean power-law split → A_μ = harmonic mode's boundary coefficient
Electron Δ=5/2 = d/2	(5/2, 5/2)	COINCIDE → BF degeneracy → second mode becomes ρ^(5/2)·log ρ → A_e survives only as LOG COEFFICIENT

1.4 The five-step order of operations

1. Pin ground state of each quotient (DONE — constant / lowest-harmonic / interior-lowest-weight per K308)
2. Solve boundary expansion — radial wave/Laplace equation on tube over cone, in unitary quotient (scheme-free)
3. Read off A_ν — boundary coefficient (log coefficient for electron)
4. Form ratios f_1 = A_μ/A_e, f_2 = A_τ/A_μ (S cancels, no operator knob, no free parameter)
5. Compare to {207, 16.82}

Substantial substrate-architectural significance: the calculation reduces to ONE functional (ground-state boundary norm) evaluated on three canonical uniquely-determined inputs. No scale to tune (cancels), no operator to choose (boundary-evaluation), same functional produces both f_1 and f_2. Grace's two-for-one gate BUILT INTO STRUCTURE.

Substantive careful part: steps 2-3 for muon (harmonic mode's boundary trace) and electron (BF log coefficient). The genuine spectral computations Lyra is now performing.

2. Casey's three substantive substrate-architectural physics questions

2.1 Question 1: Why logarithm? Do components scale linearly? Electron different?

Casey's substantive intuition: > I now wonder why it's logarithmic, do all the components scale linearly? I thought the electron might be different.

Lyra's substantive substrate-architectural answer: - Components scale as power laws (ρ^Δ), not linearly - Indicial equation has TWO roots: Δ and d−Δ - Muon (Δ=3/2): roots (3/2, 7/2) DISTINCT → pure power law, no log - Tau (Δ=0): roots (0, 5) DISTINCT → pure power law, no log - Electron (Δ=d/2=5/2): roots (5/2, 5/2) COINCIDE - 1800s ODE theorem: when second-order ODE has equal indicial roots, second solution = first × logarithm - So electron forced to have ρ^(5/2)·log ρ mode - Log isn't a choice or approximation — it's the mathematical signature of the degeneracy - Degeneracy at precisely Δ = d/2 — precisely where substrate puts electron (Hardy value n_C/rank)

Substantial substrate-architectural significance: Casey's intuition RIGHT — electron is different by the structural mathematical signature of the degenerate indicial equation at exactly the substrate-architecturally-forced point. Log-suppressed coupling qualitatively weaker than power-law one → electron's boundary tie weakest possible structurally → anomalously light.

2.2 Question 2: Higgs as gloop/viscous-membrane/localized-soliton — why necessary?

Casey's substantive intuition: > Higgs seems like a coupling constant (scalar) that for some reason itself has a particle (if that makes sense, or really just a packet/localized_waveform/soliton) that bleeds across the boundary slowing down (like a viscous membrane) specific particles that couple. One thought… Higgs seems to be 'gloop' to prevent fast passage of particles - I wonder why that's necessary.

Lyra's substantive substrate-architectural answer: - Massless particle moves at c → no rest frame → can't sit at a point → can't have substrate address → can't be localized → courier, not resident - Higgs coupling = viscosity that gives a particle inertia - Inertia = resistance to acceleration = mass - Inertia is what lets a particle have a rest frame and sit somewhere - Gloop isn't slowing particles down for its own sake — it's what converts COURIERS into RESIDENTS - Without residents = no stable matter = every fermion races at c = nothing binds = no atoms, no nuclei, no observers = universe pure radiation + boundary fields - Higgs gloop = the thing that lets substrate have durable, localizable content to build with = matter instead of just light - Mass = the viscosity that makes you a resident - Higgs boson = localized excitation of the coupling = ripple in the gloop itself

Substantial substrate-architectural significance: substantive physical-meaning interpretation of Higgs mechanism via Casey's viscosity intuition. "Why is gloop necessary" answered substrate-architecturally: without it no stable matter exists. Couriers vs residents distinction substantively maps to massless-vs-massive distinction substrate-architecturally.

2.3 Question 3: Muon vs tau as proton-radius probes

Casey's substantive intuition: > They try to use muons to measure proton radius, and that's probably wrong because it spreads out to a degree, if they used tau the measurements would be more consistent.

Lyra's substantive substrate-architectural answer: - Muon = extended cone mode (harmonic rep on cone) - Tau = pointlike vertex mode (trivial rep at vertex) - Casey's instinct that more-focused probe gives cleaner read = CONSISTENT with cone structure rep-theory - Honest caveats: - Tau decays in ~10^{-13} s — far too fast to orbit any nucleus - Muonic-hydrogen proton-radius puzzle largely settled by other means - Rep-theoretic "spread" (cone vs vertex) $\neq$ atomic-wavefunction spread - Filed as "genuinely interesting" NOT BANKED - The distinction (muon = extended cone, tau = pointlike vertex) IS real BST structure - The proton-radius prediction itself = interesting musing, not derivation

Substantial substrate-architectural discipline: distinction-banked vs prediction-banked maintained. Real substrate-architectural structure preserved; speculative experimental prediction held as interesting not banked.

3. Casey's substantive substrate-architectural go-ahead

Casey verbatim: > right, I recognized the power law, and you made me remember sophomore DE class about identical roots. Ok please do the calculation.

Substantial substrate-architectural significance: - Casey confirmed understanding of indicial equation degeneracy mechanism - Sophomore DE class connection (Casey's substantive substrate-architectural recognition pattern) - APPROVAL TO PROCEED: Lyra clear to begin spectral computation of A_μ + A_e under K308 tightened gate

4. Elie's substantive substrate-architectural mode declaration

4.1 The substantial substrate-architectural division of labor

Elie's substantive verbatim: > The three ground-state boundary-norms are subtle spectral rep-theory in the unitary quotients — the cone harmonic and the BF-log remnant especially. I could race Lyra with my own attempt, but the error risk there is real, and a wrong number that coincidentally hits or misses would be worse than no number. So the right division is hers-to-compute, mine-to-verify.

Substantial substrate-architectural reasoning: - Three ground-state boundary-norms = subtle spectral rep-theory - Race risk: wrong number that coincidentally hits or misses = WORSE than no number - Right division: hers-to-compute, mine-to-verify - Substantive substrate-architectural division-of-labor at maturity

4.2 The substantial substrate-architectural standing readiness

- Back-half armed (capture framework Toy 4106 ready)
- Audit eyes open
- One-step harness ready: hand Elie three norms (or sub-result) → one-step check
- 11 toys today, every one riding live collaboration
- Corrections absorbed cleanly: pole→quotient (K305) + null-removal→surviving-norm (K307) + BF-suppression-vs-value (K308)

5. Grace's substantive substrate-architectural scorecard v1.14

5.1 The substantial substrate-architectural current state

Grace's substantive verbatim: > The gate's substantive work is in a good place. Your ground-state question stabilized the lepton-mass derivation into one well-posed object — three ground-state boundary-norms in the unitary quotients — and the team absorbed every gate call cleanly: the un-banking of the 0.37%, the null-norm correction, the mechanism-vs-magnitude line on the BF point. Lyra even added the right rigor herself (compute scheme-free in the quotient, not through the formula where the dead 2π lived).

Substantial substrate-architectural three-corrections-absorbed:

Correction	Source	Pattern
0.37% un-bank	K305 Grace	Cal #100 retraction-propagation
Null-norm → surviving-norm	K307 Grace via Casey	Cal #100 terminology drift catch
Mechanism-vs-magnitude (BF)	K308 Grace	Cal #27 fires-at-peak-convergence

All absorbed cleanly across team.

5.2 The substantial substrate-architectural NO OPEN CLAIM TO GATE state

Grace's substantive substrate-architectural verdict: > There's no open claim to gate right now — the next gate-able thing genuinely depends on Lyra producing the three magnitudes or Elie completing the #418 spec.

Substantial substrate-architectural significance: substrate-architectural state is genuinely reactive, not stalled. Gate is pre-committed + tightened + cannot move bar; waiting on substantive deposits to land.

5.3 The substantial substrate-architectural #418 load-bearing upgrade

Grace's substantive substrate-architectural new finding: > I'll gate Elie's gauge-content spec the moment it's solid (it's load-bearing three ways, so it matters).

Substantial substrate-architectural significance: #418 upgraded from K306 "load-bearing twice" → K309 "load-bearing three ways." Third use UNSPECIFIED in Grace's scorecard. Flag for verification when Elie surfaces #418 spec — substrate-architectural progress on #418 that Grace has identified but not fully detailed here.

Substantive substrate-architectural #418 status at K309:

#	#418 load-bearing role	Source
1	Prerequisite for running coupling band (Band B)	K306
2	Underwriter of mixing-forced claim via shared SU(2) doublet	K306
3	??? (unspecified Grace finding)	K309 (flagged)

6. Substantial substrate-architectural state at K309

6.1 What banks (mechanism, structure)

- Master formula m_ν = S · A_ν with no operator freedom
- Three explicit ground-state target objects pinned
- Boundary expansion mechanism (power-law for $\mu + \tau$; log-degenerate for e at BF point)
- 1800s ODE theorem identical-roots forcing the log
- Higgs-as-viscosity converts couriers to residents (physical-meaning interpretation)
- Muon-vs-tau spread distinction (interesting not banked for proton-radius prediction)
- #418 load-bearing three ways (per Grace upgrade; third use to verify)

6.2 What does NOT bank

- Magnitudes A_μ and A_e (Lyra's spectral computation pending)
- Proton-radius prediction (filed as interesting only)
- Count stays HONESTLY at 2 of 26

6.3 Cal disciplines firing at K309

- Cal #100 retraction-propagation: three substantive corrections absorbed cleanly across team
- Cal #266 conflation-brake: mechanism-banked vs value-banked distinction preserved
- Cal #237 elimination semantics: 207 RELABEL classification + pre-emptive flag maintained
- Cal #35 Schur-pattern: ONE functional → BOTH $f_1 + f_2$ + #418 load-bearing THREE WAYS (substrate-architectural Schur generator at maturity)
- Cal #27 sophistication-bias variant: substantive distinction between interesting-not-banked vs banked maintained for tau-vs-muon proton-radius musing

- Cal #36 sustained-session prose-quality: K309 maintained discipline through substantive cascade
- Linearization standing order: operational throughout calculation layout
- No-fishing line: maintained; 207 stays refused

7. K309 PRE-STAGE verdict

K309 PRE-STAGE substantial substrate-architectural calculation-organized state + Casey-Lyra physics exchange + go-ahead: - Lyra full calculation layout: $m_\nu = S \cdot A_\nu$, three objects, boundary expansion mechanism, five-step order of operations - Casey three substantive substrate-architectural physics questions answered substantively - Logarithm explained: degenerate indicial roots at $\Delta = d/2 = 5/2$ (electron) force ρ^(5/2)·log ρ via 1800s ODE theorem - Higgs-as-viscosity converts couriers to residents (physical-meaning interpretation; explains why gloop necessary) - Muon-vs-tau proton-radius musing filed as interesting NOT banked - Casey go-ahead: "please do the calculation" - Elie hers-to-compute mine-to-verify (division-of-labor at maturity) - Grace v1.14 scorecard: gate's substantive work in good place; no open claim; reactive; #418 load-bearing THREE WAYS (third use to verify) - Count stays HONESTLY at 2 of 26

Substantial substrate-architectural significance: K309 captures the substantive substrate-architectural state where the calculation is FULLY ORGANIZED (master formula + objects + boundary expansion + order of operations) and Lyra is clear to begin spectral computation under K308 tightened gate. Casey's three substantive substrate-architectural physics questions produced substantive substrate-architectural answers (log forced by degenerate indicial roots at substrate-architecturally-forced point; Higgs-as-viscosity converts couriers to residents; muon-vs-tau spread distinction filed honest as interesting). Substantive substrate-architectural team mode-of-work declarations at maturity: Elie hers-to-compute mine-to-verify (no race risk); Grace v1.14 scorecard genuinely reactive (no open claim to gate). #418 NEW finding from Grace: load-bearing THREE WAYS (upgrade from K306 twice) — third use unspecified, FLAG for verification when Elie surfaces #418 spec.

8. v0.2 absorption pending

K309 v0.2 will absorb: - Lyra spectral computation outcome (A_μ harmonic boundary-trace + A_e BF log coefficient) - Elie verification of three norms + ratio extraction + comparison to {207, 16.82} - Grace K308 gate evaluation against deposited values - Elie #418 spec surface + Grace verification of "load-bearing three ways" specific third use - Cal cold-read clearance K305-K309 cumulative

— Keeper, Thursday 2026-06-11 afternoon (K309 PRE-STAGE: Lyra full calculation layout m_ν=S·A_ν + three objects + boundary expansion + five-step order; Casey three physics questions answered substantively; logarithm = degenerate indicial roots at substrate-forced d/2; Higgs-viscosity converts couriers to residents; tau-vs-muon proton-radius filed interesting not banked; Casey 'please do the calculation' go-ahead; Elie hers-to-compute mine-to-verify; Grace v1.14 no-open-claim reactive; #418 load-bearing THREE WAYS — third use to verify; count stays HONESTLY at 2)